Fatma Elzahraa Serry

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EDUCATION

Cairo University - Faculty of Computers and Artificial Intelligence

Bachelor of Science in Computer Science - 3.44/4 — Sep 2021 - Jun 2025

EXPERIENCE

Pixelogic Media

Software Development Engineer — Oct 2025 - Present

- Developed and maintained **front-end** features for a **web-based media authoring platform** used in subtitle and caption localization workflows.
- Refactored legacy code into modular, reusable components, which improved maintainability and shortened the time to build new features.
- Diagnosed and resolved production issues across the **rich-text editor**, **data model**, and **serialization layers**.
- Collaborated with product, QA, and fellow engineers in an Agile team, taking part in code reviews, sprint planning, and **CI/CD** releases.
- **Tech Stack**: React, Sass, Storybook, TypeScript, Vite.

Tutoruu

Software Developer — Jan 2025 - Aug 2025

- Delivered the latest release (v3) of Tutoruu, an AI-powered tutoring platform for university students, using **Next.js** and **Tailwind CSS**, resulting in a **20%** increase in user engagement through improved responsiveness and user experience.
- Developed an **AI-powered educational chatbot** using **React**, **Vite**, and **Tailwind CSS** for a client serving over **1,000** students. Integrated a biology-trained **LLM API** with real-time response handling to enable instant Q&A experiences.
- **Tech Stack**: Next.js, React Native, React, TypeScript, JavaScript, Tailwind CSS, Vite, JWT

PROJECTS

Social App — Oct 2026

- Built a **social networking app** with authenticated, protected routes and profile pages for discovering and following users.
- Integrated **REST APIs** for posts, profiles, followers, and notifications; added unread notification counts and follower/following lists.
- Used **TanStack Query** for data fetching, caching, infinite scrolling, optimistic updates, and error recovery.
- **Tech Stack**: React, TypeScript, Tailwind CSS, Vite, TanStack Query, HeroUI

SummARai - Arabic Book Summarization (Graduation Project) — Jun 2025

- Led the development of a full-stack platform for summarizing Arabic books using a hybrid **NLP pipeline** (graph-based algorithm + **AraT5** text-to-text model), with **LLaMA-4** refinement for enhanced coherence.
- Fine-tuned the AraT5v2-base-1024 model on our dataset SummARai v1.0, achieving **77.55% BERTScore Precision** in **abstractive** summarization, and deployed it on Hugging Face.
- Ranked among the **top 8** projects by the Faculty of Computers and Artificial Intelligence, Cairo University.
- Co-authoring a **research paper** on the project for submission to the **ICICIS conference,** published on **IEEE Xplore**.
- **Tech Stack**: React, Tailwind CSS, Spring Boot, FastAPI, Python, MySQL, Redis, Transformers

EXTRACURRICULAR ACTIVITIES

180 Daraga - Technical Vice Head

- Led 2024 **web development** training, mentoring members, assigning tasks, and providing feedback.
- Taught **C++ fundamentals** to over **15 students** during the Start Course v23 event.
- Served as **technical coordinator** for New Stage 16 event at Ain Shams University, implementing **backend endpoints** for registration, interviews, and email communication, supporting **900+** participants.

SKILLS

- **Technologies & Tools:** React, Next.js, Node.js, MongoDB, Sass, Tailwind CSS, Git
- **Languages:** JavaScript, TypeScript, Java, Python, C++, SQL
- **Concepts:** Agile Methodology, Data Structures and Algorithms, Problem Solving, Design Patterns, Object-Oriented Programming (OOP), SOLID Principles, RESTful APIs